Table 1 Target pairs and experimental activity classes

Target pair	PDB A	PDB B	Dual (n)	A-only (n)	B-only (n)	Neither (n)	Total (n)
EGFR/HER2	3POZ	3RCD	28	37	32	12	109
JAK1/JAK2	6N7A	8BXH	32	32	32	14	110
JAK1/TYK2	6N7A	3LXP	32	32	32	14	110
PIK3CA/mTOR	4L23	4JT6	32	32	32	14	110
AChE/BChE	4EY7	4BDS	28	27	28	15	98
F2/F10	4UDW	2JKH	32	32	32	14	110
PPARG/PPARA	9V8H	6LXA	32	32	32	14	110
PPARA/PPARD	6LXA	5U3Q	32	32	32	14	110

Counts are target-pair–ligand observations; the 867 observations represent 845 distinct ligand entities. Target A and target B follow the order in each pair name. Activity is defined by pChEMBL ≥ 6.0. PDB identifiers refer to the receptors used in the primary analysis.

Table 2 Dual-ligand recovery in matched top-10% candidate lists

Target pair	Scheme	N	k	Dual selected S1 → scheme	Dual recall (%) S1 → scheme
EGFR/HER2	S2	97	10	1 → 2	3.57 → 7.14
EGFR/HER2	S3	97	10	1 → 2	3.57 → 7.14
EGFR/HER2	S4[a]	97	10	1 → 2	3.57 → 7.14
EGFR/HER2	S5	97	10	1 → 2	3.57 → 7.14
JAK1/JAK2	S2	96	10	6 → 7	18.75 → 21.88
JAK1/JAK2	S3	96	10	6 → 4	18.75 → 12.50
JAK1/JAK2	S4[a]	96	10	6 → 4	18.75 → 12.50
JAK1/JAK2	S5	96	10	6 → 7	18.75 → 21.88
JAK1/TYK2	S2	96	10	3 → 6	9.38 → 18.75
JAK1/TYK2	S3	96	10	3 → 5	9.38 → 15.63
JAK1/TYK2	S4[a]	96	10	3 → 5	9.38 → 15.63
JAK1/TYK2	S5	96	10	3 → 5	9.38 → 15.63
PIK3CA/mTOR	S2	96	10	4 → 6	12.50 → 18.75
PIK3CA/mTOR	S3	96	10	4 → 7	12.50 → 21.88
PIK3CA/mTOR	S4[a]	96	10	4 → 5	12.50 → 15.63
PIK3CA/mTOR	S5	96	10	4 → 7	12.50 → 21.88
AChE/BChE	S2	79	8	6 → 3	22.22 → 11.11
AChE/BChE	S3	80	8	6 → 3	22.22 → 11.11
AChE/BChE	S4[a]	79	8	6 → 3	22.22 → 11.11
AChE/BChE	S5	80	8	6 → 3	22.22 → 11.11
F2/F10	S2	95	10	4 → 4	12.90 → 12.90
F2/F10	S3	96	10	4 → 7	12.50 → 21.88
F2/F10	S4[a]	95	10	4 → 5	12.90 → 16.13
F2/F10	S5	95	10	4 → 6	12.90 → 19.35
PPARG/PPARA	S2	94	10	6 → 2	20.00 → 6.67
PPARG/PPARA	S3	96	10	6 → 4	18.75 → 12.50
PPARG/PPARA	S4[a]	94	10	6 → 3	20.00 → 10.00
PPARG/PPARA	S5	96	10	6 → 1	18.75 → 3.13
PPARA/PPARD	S2	91	10	3 → 5	11.11 → 18.52
PPARA/PPARD	S3	96	10	3 → 6	9.38 → 18.75
PPARA/PPARD	S4[a]	91	10	3 → 4	11.11 → 14.81
PPARA/PPARD	S5	96	10	3 → 5	9.38 → 15.63

Each row uses its own common two-target complete-case population (N); k = ceil(0.10 N). Arrows show S1 → comparison scheme. Dual recall (%) is the fraction of all dual ligands recovered. Counts and recall retain comparison-specific S1 baselines. Full class composition: Table S10.

[a] Supporting comparison against S1.

Table S1 Evaluation-set construction and activity provenance

A. Selection conditions

Target pair	Selection threshold	Chemical selection record	Deposited quota / composition	Eligible n	Selection replay status
EGFR/HER2	6	Murcko cap 5 within class	deposited N=110 with classes 28/38/32/12 before the later unresolved-arm exclusion	109	Documented; not replayed
JAK1/JAK2	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
JAK1/TYK2	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PIK3CA/mTOR	6	Fixed 110-molecule membership	32 / 32 / 32 / 14	110	Not independently replayed in this compilation
AChE/BChE	6.5 / 5.5	ChEMBL ID-prefix diversity key in deposited script	N=100; dual / A-only / B-only / neither 28/28/28/16	98	Documented; not replayed
F2/F10	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PPARG/PPARA	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PPARA/PPARD	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed

All final labels use pChEMBL >= 6.0. Provenance flags overlap. Complete identifiers require DOI, PubMed or patent identifiers for all tied maxima at both endpoints. Across all pairs, cross-document aggregation is present in 81 observations and endpoint heterogeneity in 123 (union 168). The full source crosswalk, raw activity rows, selection evidence and exceptions are included in Supplementary_Data.zip. CAP5 is a post hoc sensitivity analysis and does not redefine initial selection.

Table S1 Evaluation-set construction and activity provenance

B. Activity-source coverage

Target pair	n	Both ends traced	Complete identifiers	Missing identifiers	Source / endpoint mix	Threshold crossing (n)	Value / label mismatch
EGFR/HER2	109	109	101	8	22	13	0
JAK1/JAK2	110	110	109	1	9	0	0
JAK1/TYK2	110	110	108	2	13	4	0
PIK3CA/mTOR	110	110	109	1	24	3	0
AChE/BChE	98	98	98	0	8	2	0
F2/F10	110	110	110	0	10	0	0
PPARG/PPARA	110	110	110	0	42	9	0
PPARA/PPARD	110	110	110	0	40	4	0

All final labels use pChEMBL >= 6.0. Provenance flags overlap. Complete identifiers require DOI, PubMed or patent identifiers for all tied maxima at both endpoints. Across all pairs, cross-document aggregation is present in 81 observations and endpoint heterogeneity in 123 (union 168). The full source crosswalk, raw activity rows, selection evidence and exceptions are included in Supplementary_Data.zip. CAP5 is a post hoc sensitivity analysis and does not redefine initial selection.

Table S2 Receptor structures and docking boxes

A. Receptor identification

Role	Target	PDB	Auth chains	Cognate CCD	Cognate instance	Retained / removed components
Primary	EGFR	3POZ	A	03P	A:1023	Selected protein chain retained
Primary	HER2	3RCD	A	03P	A:9001	Selected protein chain retained
Primary	JAK1	6N7A	AB	KEV	A:1203	Chains A and B retained; PTR converted to TYR
Primary	JAK2	8BXH	A	C87	A:1201	PTR converted to TYR
Primary	TYK2	3LXP	A	IZA	A:1183	PTR converted to TYR
Primary	PIK3CA	4L23	A	X6K	A:1101	p85 removed
Primary	mTOR	4JT6	A	X6K	A:2601	Second mTOR copy and mLST8 removed
Primary	AChE	4EY7	AB	E20	A:604	AChE chains A and B retained
Primary	BChE	4BDS	A	THA	A:701	Selected protein chain retained
Primary	F2	4UDW	HIL	N6L	H:1249	Heavy chain, light chain and hirudin retained
Primary	F10	2JKH	AL	BI7	A:1244	Heavy and light chains retained
Primary	PPARG	9V8H	AB	BRL	A:501	PPARG and PG08-NL peptide retained
Primary	PPARA	6LXA	A	EPA	A:502	Selected protein chain retained
Primary	PPARD	5U3Q	B	7UJ	B:501	PPARD chain B retained
Alternative	AChE	4EY6	AB	GNT	A:604	AChE chains A and B retained
Alternative	BChE	1P0M	A	CHT	A:607	Selected protein chain retained
Alternative	F2	3SHC	HIL	B01	H:3	Heavy chain, light chain and hirudin retained
Alternative	F10	2Y5F	AL	XWG	A:1244	Heavy and light chains retained
Alternative	PPARA	6KAX	A	PLM	A:502	Selected protein chain retained
Alternative	PPARD	5U46	A	7T1	A:505	Selected protein chain retained

Coordinates and dimensions are in angstrom. Waters and cocrystal ligands are removed from the receptor. Box centers are bounding-box midpoints; padding is 5 Å with minimum edge 20 Å. Chain identifiers are author chain identifiers. Formal prepared files and preparation exceptions, including observed-fragment templates, are documented in the accompanying source files. Alternative structures cover only the six conditions used in the current analysis; no Q110 alternative-receptor result is included.

Table S2 Receptor structures and docking boxes

B. Docking-box coordinates

Role	Target	PDB	Center (x, y, z), Å	Size (x, y, z), Å
Primary	EGFR	3POZ	18.7460, 31.8320, 11.6260	20.7500, 20.0000, 21.2180
Primary	HER2	3RCD	12.4800, 2.9635, 28.0150	21.1620, 22.1630, 20.0000
Primary	JAK1	6N7A	5.9110, -24.3175, 21.4945	20.0000, 20.5970, 20.0000
Primary	JAK2	8BXH	-12.5897, 14.5939, 21.2626	24.0687, 20.0000, 20.0000
Primary	TYK2	3LXP	-4.4985, 25.5775, -31.0640	20.0000, 20.0000, 20.0000
Primary	PIK3CA	4L23	32.5530, 45.3340, 42.0935	20.0000, 20.0000, 20.0000
Primary	mTOR	4JT6	52.1230, -0.0840, -47.6645	20.4680, 20.0000, 20.0000
Primary	AChE	4EY7	-13.9875, -43.9055, 27.1085	23.3410, 20.0000, 20.3550
Primary	BChE	4BDS	133.0760, 116.1130, 41.3350	20.0000, 20.0000, 20.0000
Primary	F2	4UDW	129.7390, -14.4025, 164.9460	20.6000, 20.0000, 20.0000
Primary	F10	2JKH	-3.6430, 10.7695, 22.7710	20.0000, 20.0000, 21.1180
Primary	PPARG	9V8H	-5.4540, -5.3730, -23.5915	20.0000, 20.1400, 20.0000
Primary	PPARA	6LXA	11.9975, 5.7420, -7.4430	20.0000, 20.2000, 23.4320
Primary	PPARD	5U3Q	45.4160, -37.6795, 103.9765	20.0000, 20.0000, 22.4490
Alternative	AChE	4EY6	-10.4920, -43.5255, 29.4535	20, 20, 20
Alternative	BChE	1P0M	133.5840, 116.7965, 40.3950	20, 20, 20
Alternative	F2	3SHC	16.7620, -12.0230, 22.9065	20.5680, 20, 20
Alternative	F10	2Y5F	-4.6240, 10.6510, 22.4765	20, 20.2300, 21.4290
Alternative	PPARA	6KAX	12.0005, 6.0880, -7.6585	20, 20, 22.6890
Alternative	PPARD	5U46	45.4215, -1.7525, 135.5685	20, 20, 20.4870

Coordinates and dimensions are in angstrom. Waters and cocrystal ligands are removed from the receptor. Box centers are bounding-box midpoints; padding is 5 Å with minimum edge 20 Å. Chain identifiers are author chain identifiers. Formal prepared files and preparation exceptions, including observed-fragment templates, are documented in the accompanying source files. Alternative structures cover only the six conditions used in the current analysis; no Q110 alternative-receptor result is included.

Table S3 Cocrystal-ligand redocking across five random seeds

Target	PDB	CCD	Seeds	Jobs	Top-1 RMSD median [min, max] (Å)	Best-saved RMSD median [min, max] (Å)	Top-1 <2 Å (n)	Best <2 Å (n)	Modes
EGFR	3POZ	03P	5	5	1.0960 [1.0940, 1.1170]	1.0960 [1.0940, 1.1170]	5	5	9
HER2	3RCD	03P	5	5	2.0540 [2.0210, 2.8800]	2.0460 [1.3370, 2.0580]	0	1	9
JAK1	6N7A	KEV	5	5	0.4560 [0.4440, 0.4740]	0.4560 [0.4440, 0.4740]	5	5	9
JAK2	8BXH	C87	5	5	10.6940 [10.6860, 10.7200]	0.9980 [0.9490, 1.0750]	0	5	9
TYK2	3LXP	IZA	5	5	0.2560 [0.2180, 0.2620]	0.2560 [0.2180, 0.2620]	5	5	9
PIK3CA	4L23	X6K	5	5	0.9680 [0.8960, 0.9790]	0.9680 [0.8960, 0.9790]	5	5	9
mTOR	4JT6	X6K	5	5	4.9280 [4.9270, 4.9330]	0.5810 [0.5550, 0.6010]	0	5	9
AChE	4EY7	E20	5	5	0.8030 [0.7870, 0.8280]	0.8030 [0.7870, 0.8280]	5	5	9
BChE	4BDS	THA	5	5	4.7540 [4.7530, 4.7540]	0.3140 [0.3040, 0.4290]	0	5	9
F2	4UDW	N6L	5	5	0.5380 [0.5270, 0.5680]	0.5380 [0.5270, 0.5680]	5	5	9
F10	2JKH	BI7	5	5	0.8220 [0.7790, 0.8480]	0.8220 [0.7790, 0.8480]	5	5	9
PPARG	9V8H	BRL	5	5	2.0810 [1.6560, 6.7880]	1.4400 [0.7280, 1.9770]	2	5	9
PPARA	6LXA	EPA	5	5	7.7870 [7.7100, 7.8410]	1.1250 [0.5970, 1.6120]	0	5	9
PPARD	5U3Q	7UJ	5	5	0.4840 [0.4460, 0.7780]	0.4840 [0.4460, 0.7780]	5	5	9

Seeds: 17, 29, 42, 71 and 101. Symmetry-aware heavy-atom RMSD is calculated in the receptor coordinate frame. Top-ranked-pose success and success among saved poses are separate outcomes. The summary uses recorded RMSDs without recomputing coordinates; every seed-level record is in Supplementary_Data.zip. CCD denotes the cognate ligand Chemical Component Dictionary identifier; jobs denotes successful docking jobs; modes denotes saved poses per job.

Table S4 Analysis populations and score availability

A. Directional and chemistry populations

Target pair	Direction	Labeled n	S1 directional n	S1 missing n	Unassigned fold after S1	Chemistry OOF n
EGFR/HER2	Dual vs A-only	65	65	0	0	65
EGFR/HER2	Dual vs B-only	60	60	0	1	59
JAK1/JAK2	Dual vs A-only	64	64	0	0	64
JAK1/JAK2	Dual vs B-only	64	64	0	0	64
JAK1/TYK2	Dual vs A-only	64	64	0	1	63
JAK1/TYK2	Dual vs B-only	64	64	0	1	63
PIK3CA/mTOR	Dual vs A-only	64	64	0	0	64
PIK3CA/mTOR	Dual vs B-only	64	64	0	0	64
AChE/BChE	Dual vs A-only	55	52	3	0	52
AChE/BChE	Dual vs B-only	56	55	1	0	55
F2/F10	Dual vs A-only	64	64	0	1	63
F2/F10	Dual vs B-only	64	64	0	1	63
PPARG/PPARA	Dual vs A-only	64	64	0	1	63
PPARG/PPARA	Dual vs B-only	64	64	0	0	64
PPARA/PPARD	Dual vs A-only	64	64	0	0	64
PPARA/PPARD	Dual vs B-only	64	64	0	0	64

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S5B.

Table S4 Analysis populations and score availability

B. Score availability across activity classes

Target pair	Scheme	Expected A	Valid A	Missing A	Expected B	Valid B	Missing B
EGFR/HER2	S1	109	109	0	109	109	0
EGFR/HER2	S2	109	109	0	109	109	0
EGFR/HER2	S3	109	109	0	109	109	0
EGFR/HER2	S4	109	109	0	109	109	0
EGFR/HER2	S5	109	109	0	109	109	0
JAK1/JAK2	S1	110	110	0	110	110	0
JAK1/JAK2	S2	110	110	0	110	110	0
JAK1/JAK2	S3	110	110	0	110	110	0
JAK1/JAK2	S4	110	110	0	110	110	0
JAK1/JAK2	S5	110	110	0	110	110	0
JAK1/TYK2	S1	110	110	0	110	110	0
JAK1/TYK2	S2	110	110	0	110	110	0
JAK1/TYK2	S3	110	110	0	110	110	0
JAK1/TYK2	S4	110	110	0	110	110	0
JAK1/TYK2	S5	110	110	0	110	110	0
PIK3CA/mTOR	S1	110	110	0	110	110	0
PIK3CA/mTOR	S2	110	110	0	110	110	0
PIK3CA/mTOR	S3	110	110	0	110	110	0
PIK3CA/mTOR	S4	110	110	0	110	110	0
PIK3CA/mTOR	S5	110	110	0	110	110	0
AChE/BChE	S1	98	95	3	98	95	3
AChE/BChE	S2	98	94	4	98	94	4
AChE/BChE	S3	98	95	3	98	95	3
AChE/BChE	S4	98	94	4	98	94	4
AChE/BChE	S5	98	97	1	98	97	1
F2/F10	S1	110	110	0	110	110	0
F2/F10	S2	110	109	1	110	110	0
F2/F10	S3	110	110	0	110	110	0
F2/F10	S4	110	109	1	110	110	0
F2/F10	S5	110	108	2	110	108	2

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S5B.

Table S4 Analysis populations and score availability (continued)

B. Score availability across activity classes

Target pair	Scheme	Expected A	Valid A	Missing A	Expected B	Valid B	Missing B
PPARG/PPARA	S1	110	110	0	110	110	0
PPARG/PPARA	S2	110	108	2	110	108	2
PPARG/PPARA	S3	110	110	0	110	110	0
PPARG/PPARA	S4	110	108	2	110	108	2
PPARG/PPARA	S5	110	110	0	110	110	0
PPARA/PPARD	S1	110	110	0	110	110	0
PPARA/PPARD	S2	110	105	5	110	105	5
PPARA/PPARD	S3	110	110	0	110	110	0
PPARA/PPARD	S4	110	105	5	110	105	5
PPARA/PPARD	S5	110	110	0	110	110	0

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S5B.

Table S5 Directional discrimination and paired scoring-workflow comparisons

A. Method-specific estimates

Target pair	Scheme	n (dual/A/B)	AUROC B [95% CI]	AUROC A [95% CI]	Min. AUROC [95% CI]
EGFR/HER2	S1	28 / 37 / 32	0.6274 [0.4865, 0.7645]	0.2467 [0.1328, 0.3783]	0.2467 [0.1328, 0.3783]
EGFR/HER2	S2	28 / 37 / 32	0.3731 [0.2355, 0.5150]	0.2210 [0.0993, 0.3538]	0.2210 [0.0993, 0.3516]
EGFR/HER2	S3	28 / 37 / 32	0.5241 [0.3793, 0.6641]	0.2634 [0.1339, 0.4051]	0.2634 [0.1339, 0.4051]
EGFR/HER2	S4	28 / 37 / 32	0.4204 [0.2760, 0.5690]	0.2812 [0.1551, 0.4208]	0.2812 [0.1551, 0.4163]
EGFR/HER2	S5	28 / 37 / 32	0.4344 [0.2896, 0.5820]	0.2455 [0.1239, 0.3806]	0.2455 [0.1239, 0.3806]
JAK1/JAK2	S1	32 / 32 / 32	0.5908 [0.4512, 0.7275]	0.7485 [0.6235, 0.8604]	0.5908 [0.4512, 0.7246]
JAK1/JAK2	S2	32 / 32 / 32	0.4385 [0.2979, 0.5791]	0.7402 [0.6055, 0.8623]	0.4385 [0.2979, 0.5791]
JAK1/JAK2	S3	32 / 32 / 32	0.4736 [0.3291, 0.6211]	0.7480 [0.6162, 0.8623]	0.4736 [0.3291, 0.6201]
JAK1/JAK2	S4	32 / 32 / 32	0.4443 [0.2988, 0.5898]	0.6846 [0.5430, 0.8184]	0.4443 [0.2988, 0.5898]
JAK1/JAK2	S5	32 / 32 / 32	0.5137 [0.3682, 0.6592]	0.7012 [0.5605, 0.8311]	0.5137 [0.3682, 0.6592]
JAK1/TYK2	S1	32 / 32 / 32	0.6475 [0.5049, 0.7812]	0.4219 [0.2812, 0.5654]	0.4219 [0.2812, 0.5635]
JAK1/TYK2	S2	32 / 32 / 32	0.6631 [0.5225, 0.7940]	0.5596 [0.4102, 0.7021]	0.5596 [0.4102, 0.6836]
JAK1/TYK2	S3	32 / 32 / 32	0.7090 [0.5742, 0.8271]	0.4873 [0.3379, 0.6338]	0.4873 [0.3379, 0.6309]
JAK1/TYK2	S4	32 / 32 / 32	0.7637 [0.6348, 0.8779]	0.6758 [0.5254, 0.8164]	0.6758 [0.5254, 0.7998]
JAK1/TYK2	S5	32 / 32 / 32	0.6689 [0.5322, 0.8067]	0.6846 [0.5430, 0.8115]	0.6689 [0.5176, 0.7666]
PIK3CA/mTOR	S1	32 / 32 / 32	0.7363 [0.6074, 0.8486]	0.5801 [0.4346, 0.7188]	0.5801 [0.4346, 0.7129]
PIK3CA/mTOR	S2	32 / 32 / 32	0.5752 [0.4297, 0.7168]	0.6592 [0.5195, 0.7871]	0.5752 [0.4287, 0.6982]
PIK3CA/mTOR	S3	32 / 32 / 32	0.7041 [0.5674, 0.8281]	0.7305 [0.5986, 0.8467]	0.7041 [0.5596, 0.7998]
PIK3CA/mTOR	S4	32 / 32 / 32	0.5166 [0.3730, 0.6602]	0.6641 [0.5263, 0.7910]	0.5166 [0.3730, 0.6523]
PIK3CA/mTOR	S5	32 / 32 / 32	0.5977 [0.4492, 0.7344]	0.7148 [0.5801, 0.8350]	0.5977 [0.4482, 0.7285]
AChE/BChE	S1	27 / 25 / 28	0.6341 [0.4756, 0.7867]	0.6442 [0.4907, 0.7871]	0.6341 [0.4607, 0.7540]
AChE/BChE	S2	27 / 25 / 27	0.5052 [0.3392, 0.6667]	0.4925 [0.3333, 0.6502]	0.4925 [0.3141, 0.5867]
AChE/BChE	S3	27 / 25 / 28	0.6474 [0.4903, 0.7881]	0.4907 [0.3360, 0.6468]	0.4907 [0.3360, 0.6296]
AChE/BChE	S4	27 / 25 / 27	0.5437 [0.3822, 0.6978]	0.3909 [0.2401, 0.5528]	0.3909 [0.2401, 0.5336]
AChE/BChE	S5	27 / 27 / 28	0.6228 [0.4650, 0.7668]	0.4511 [0.2923, 0.6111]	0.4511 [0.2923, 0.5967]
F2/F10	S1	32 / 32 / 32	0.4082 [0.2656, 0.5557]	0.3623 [0.2363, 0.5049]	0.3623 [0.2305, 0.4805]
F2/F10	S2	31 / 32 / 32	0.5383 [0.3911, 0.6774]	0.5675 [0.4194, 0.7067]	0.5383 [0.3821, 0.6300]
F2/F10	S3	32 / 32 / 32	0.6094 [0.4668, 0.7490]	0.6230 [0.4785, 0.7568]	0.6094 [0.4531, 0.7021]
F2/F10	S4	31 / 32 / 32	0.6341 [0.4919, 0.7641]	0.5696 [0.4173, 0.7127]	0.5696 [0.4163, 0.6855]
F2/F10	S5	31 / 32 / 32	0.5625 [0.4133, 0.7107]	0.5554 [0.4062, 0.7026]	0.5554 [0.3921, 0.6633]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S10 has no added CI.

Table S5 Directional discrimination and paired scoring-workflow comparisons (continued)

A. Method-specific estimates

Target pair	Scheme	n (dual/A/B)	AUROC B [95% CI]	AUROC A [95% CI]	Min. AUROC [95% CI]
PPARG/PPARA	S1	32 / 32 / 32	0.6553 [0.5117, 0.7861]	0.6021 [0.4614, 0.7402]	0.6021 [0.4561, 0.7051]
PPARG/PPARA	S2	30 / 32 / 32	0.6583 [0.5177, 0.7885]	0.2802 [0.1573, 0.4156]	0.2802 [0.1573, 0.4156]
PPARG/PPARA	S3	32 / 32 / 32	0.6455 [0.5029, 0.7744]	0.3105 [0.1748, 0.4492]	0.3105 [0.1748, 0.4492]
PPARG/PPARA	S4	30 / 32 / 32	0.5781 [0.4313, 0.7229]	0.3312 [0.2000, 0.4719]	0.3312 [0.2000, 0.4698]
PPARG/PPARA	S5	32 / 32 / 32	0.5146 [0.3701, 0.6543]	0.4463 [0.3066, 0.5957]	0.4463 [0.3037, 0.5596]
PPARA/PPARD	S1	32 / 32 / 32	0.6787 [0.5430, 0.8047]	0.5107 [0.3652, 0.6543]	0.5107 [0.3652, 0.6475]
PPARA/PPARD	S2	27 / 32 / 32	0.7002 [0.5590, 0.8287]	0.6817 [0.5417, 0.8113]	0.6817 [0.5255, 0.7546]
PPARA/PPARD	S3	32 / 32 / 32	0.7520 [0.6240, 0.8653]	0.6689 [0.5332, 0.7979]	0.6689 [0.5322, 0.7725]
PPARA/PPARD	S4	27 / 32 / 32	0.7581 [0.6296, 0.8738]	0.5625 [0.4144, 0.7118]	0.5625 [0.4144, 0.7025]
PPARA/PPARD	S5	32 / 32 / 32	0.6338 [0.4922, 0.7676]	0.6670 [0.5264, 0.7998]	0.6338 [0.4844, 0.7197]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S10 has no added CI.

Table S5 Directional discrimination and paired scoring-workflow comparisons

B. Paired differences in minimum AUROC

Target pair	Contrast	n (dual/A/B)	Comparison minimum AUROC	Reference minimum AUROC	Difference (95% CI)
EGFR/HER2	S2-S1	28 / 37 / 32	0.2210	0.2467	-0.0257 [-0.1775, 0.1283]
EGFR/HER2	S3-S1	28 / 37 / 32	0.2634	0.2467	0.0167 [-0.1340, 0.1685]
EGFR/HER2	S4-S2	28 / 37 / 32	0.2812	0.2210	0.0603 [-0.0536, 0.1797]
EGFR/HER2	S5-S1	28 / 37 / 32	0.2455	0.2467	-0.0011 [-0.1629, 0.1596]
JAK1/JAK2	S2-S1	32 / 32 / 32	0.4385	0.5908	-0.1523 [-0.3672, 0.0605]
JAK1/JAK2	S3-S1	32 / 32 / 32	0.4736	0.5908	-0.1172 [-0.2861, 0.0498]
JAK1/JAK2	S4-S2	32 / 32 / 32	0.4443	0.4385	0.0059 [-0.1328, 0.1406]
JAK1/JAK2	S5-S1	32 / 32 / 32	0.5137	0.5908	-0.0771 [-0.2920, 0.1396]
JAK1/TYK2	S2-S1	32 / 32 / 32	0.5596	0.4219	0.1377 [-0.0371, 0.2969]
JAK1/TYK2	S3-S1	32 / 32 / 32	0.4873	0.4219	0.0654 [-0.0791, 0.2109]
JAK1/TYK2	S4-S2	32 / 32 / 32	0.6758	0.5596	0.1162 [-0.0127, 0.2539]
JAK1/TYK2	S5-S1	32 / 32 / 32	0.6689	0.4219	0.2471 [0.0312, 0.4082]
PIK3CA/mTOR	S2-S1	32 / 32 / 32	0.5752	0.5801	-0.0049 [-0.1924, 0.1680]
PIK3CA/mTOR	S3-S1	32 / 32 / 32	0.7041	0.5801	0.1240 [-0.0469, 0.2607]
PIK3CA/mTOR	S4-S2	32 / 32 / 32	0.5166	0.5752	-0.0586 [-0.1660, 0.0547]
PIK3CA/mTOR	S5-S1	32 / 32 / 32	0.5977	0.5801	0.0176 [-0.1807, 0.2178]
AChE/BChE	S2-S1	27 / 25 / 27	0.4925	0.6341	-0.1416 [-0.3676, 0.0489]
AChE/BChE	S3-S1	27 / 25 / 28	0.4907	0.6341	-0.1433 [-0.2608, 0.0172]
AChE/BChE	S4-S2	27 / 25 / 27	0.3909	0.4925	-0.1015 [-0.2215, 0.0923]
AChE/BChE	S5-S1	27 / 25 / 28	0.4511	0.6341	-0.1830 [-0.4026, 0.0715]
F2/F10	S2-S1	31 / 32 / 32	0.5383	0.3740	0.1643 [-0.0141, 0.3054]
F2/F10	S3-S1	32 / 32 / 32	0.6094	0.3623	0.2471 [0.0654, 0.3896]
F2/F10	S4-S2	31 / 32 / 32	0.5696	0.5383	0.0312 [-0.0796, 0.1724]
F2/F10	S5-S1	31 / 32 / 32	0.5554	0.3740	0.1815 [-0.0071, 0.3357]
PPARG/PPARA	S2-S1	30 / 32 / 32	0.2802	0.5891	-0.3089 [-0.4688, -0.1313]
PPARG/PPARA	S3-S1	32 / 32 / 32	0.3105	0.6021	-0.2915 [-0.4380, -0.1104]
PPARG/PPARA	S4-S2	30 / 32 / 32	0.3312	0.2802	0.0510 [-0.0719, 0.1740]
PPARG/PPARA	S5-S1	32 / 32 / 32	0.4463	0.6021	-0.1558 [-0.3457, 0.0459]
PPARA/PPARD	S2-S1	27 / 32 / 32	0.6817	0.5625	0.1192 [-0.0706, 0.2697]
PPARA/PPARD	S3-S1	32 / 32 / 32	0.6689	0.5107	0.1582 [0.0059, 0.2979]
PPARA/PPARD	S4-S2	27 / 32 / 32	0.5625	0.6817	-0.1192 [-0.2616, 0.0868]
PPARA/PPARD	S5-S1	32 / 32 / 32	0.6338	0.5107	0.1230 [-0.0762, 0.2754]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S10 has no added CI.

Table S6 Negative-class and pocket-correspondence controls

A. Negative-class replacement

Target pair	Direction	n (dual / single / neither)	AUROC single-target	AUROC neither	Difference (95% CI)
EGFR/HER2	Dual vs A-only	28 / 37 / 12	0.6274	0.7083	0.0809 [-0.1447, 0.2871]
EGFR/HER2	Dual vs B-only	28 / 32 / 12	0.2467	0.7470	0.5004 [0.3025, 0.6819]
JAK1/JAK2	Dual vs A-only	32 / 32 / 14	0.5908	0.7344	0.1436 [-0.0498, 0.3326]
JAK1/JAK2	Dual vs B-only	32 / 32 / 14	0.7485	0.7143	-0.0342 [-0.2238, 0.1410]
JAK1/TYK2	Dual vs A-only	32 / 32 / 14	0.6475	0.7321	0.0847 [-0.0979, 0.2578]
JAK1/TYK2	Dual vs B-only	32 / 32 / 14	0.4219	0.8058	0.3839 [0.1953, 0.5693]
PIK3CA/mTOR	Dual vs A-only	32 / 32 / 14	0.7363	0.7679	0.0315 [-0.0897, 0.1497]
PIK3CA/mTOR	Dual vs B-only	32 / 32 / 14	0.5801	0.6964	0.1164 [-0.0592, 0.2872]
AChE/BChE	Dual vs A-only	27 / 25 / 15	0.6341	0.6963	0.0622 [-0.1047, 0.2242]
AChE/BChE	Dual vs B-only	27 / 28 / 15	0.6442	0.5679	-0.0763 [-0.2019, 0.0469]
F2/F10	Dual vs A-only	32 / 32 / 14	0.4082	0.5045	0.0963 [-0.0466, 0.2538]
F2/F10	Dual vs B-only	32 / 32 / 14	0.3623	0.4888	0.1265 [-0.0172, 0.2755]
PPARG/PPARA	Dual vs A-only	32 / 32 / 14	0.6553	0.7567	0.1014 [-0.0724, 0.2727]
PPARG/PPARA	Dual vs B-only	32 / 32 / 14	0.6021	0.7054	0.1033 [-0.1187, 0.3121]
PPARA/PPARD	Dual vs A-only	32 / 32 / 14	0.6787	0.6518	-0.0269 [-0.2102, 0.1486]
PPARA/PPARD	Dual vs B-only	32 / 32 / 14	0.5107	0.4866	-0.0241 [-0.1915, 0.1482]

Panel A difference = neither minus single-target negative class. Panel B difference = corresponding minus noncorresponding pocket. All estimates use S1. Individual AUROC confidence intervals and ligand identifiers are retained in the complete source CSVs. Corresp., corresponding pocket; noncorresp., noncorresponding pocket. Grouped counts follow the class order in the column heading.

Table S6 Negative-class and pocket-correspondence controls

B. Corresponding and noncorresponding pockets

Target pair	Direction	n (dual / negative)	Corresp. AUROC	Noncorresp. AUROC	Difference (95% CI)	Additional exclusions
EGFR/HER2	Dual vs A-only	28 / 37	0.6274	0.6573	-0.0299 [-0.0936, 0.0271]	0
EGFR/HER2	Dual vs B-only	28 / 32	0.2467	0.2355	0.0112 [-0.0681, 0.0949]	0
JAK1/JAK2	Dual vs A-only	32 / 32	0.5908	0.5908	0 [-0.0771, 0.0703]	0
JAK1/JAK2	Dual vs B-only	32 / 32	0.7485	0.7012	0.0474 [-0.0278, 0.1338]	0
JAK1/TYK2	Dual vs A-only	32 / 32	0.6475	0.6250	0.0225 [-0.0898, 0.1397]	0
JAK1/TYK2	Dual vs B-only	32 / 32	0.4219	0.4634	-0.0415 [-0.1279, 0.0410]	0
PIK3CA/mTOR	Dual vs A-only	32 / 32	0.7363	0.6519	0.0845 [-0.0454, 0.2144]	0
PIK3CA/mTOR	Dual vs B-only	32 / 32	0.5801	0.5859	-0.0059 [-0.1230, 0.1094]	0
AChE/BChE	Dual vs A-only	27 / 25	0.6341	0.4681	0.1659 [0.0548, 0.2830]	0
AChE/BChE	Dual vs B-only	27 / 28	0.6442	0.5807	0.0635 [-0.0661, 0.1971]	0
F2/F10	Dual vs A-only	32 / 32	0.4082	0.3926	0.0156 [-0.0957, 0.1230]	0
F2/F10	Dual vs B-only	32 / 32	0.3623	0.3887	-0.0264 [-0.1465, 0.0967]	0
PPARG/PPARA	Dual vs A-only	32 / 32	0.6553	0.5928	0.0625 [-0.1035, 0.2275]	0
PPARG/PPARA	Dual vs B-only	32 / 32	0.6021	0.6909	-0.0889 [-0.2563, 0.0869]	0
PPARA/PPARD	Dual vs A-only	32 / 32	0.6787	0.4434	0.2354 [0.0693, 0.4043]	0
PPARA/PPARD	Dual vs B-only	32 / 32	0.5107	0.4561	0.0547 [-0.1436, 0.2441]	0

Panel A difference = neither minus single-target negative class. Panel B difference = corresponding minus noncorresponding pocket. All estimates use S1. Individual AUROC confidence intervals and ligand identifiers are retained in the complete source CSVs. Corresp., corresponding pocket; noncorresp., noncorresponding pocket. Grouped counts follow the class order in the column heading.

Table S7 Out-of-fold performance of ligand-chemistry models

Target pair	Direction	Common OOF n	Descriptor AUROC	ECFP4 AUROC	ECFP4 + S1 AUROC	Matched S1 AUROC	ECFP4 + S1 increment
EGFR/HER2	Dual vs A-only	65	0.6757	0.8002	0.7838	0.6274	-0.0164
EGFR/HER2	Dual vs B-only	59	0.7592	0.8790	0.8767	0.2408	-0.0023
JAK1/JAK2	Dual vs A-only	64	0.6030	0.9185	0.9199	0.5908	0.0015
JAK1/JAK2	Dual vs B-only	64	0.9014	0.9678	0.9707	0.7485	0.0029
JAK1/TYK2	Dual vs A-only	63	0.5655	0.8488	0.8438	0.6371	-0.0050
JAK1/TYK2	Dual vs B-only	63	0.8407	0.9042	0.9032	0.4032	-0.0010
PIK3CA/mTOR	Dual vs A-only	64	0.6094	0.8213	0.8428	0.7363	0.0215
PIK3CA/mTOR	Dual vs B-only	64	0.2939	0.7285	0.7256	0.5801	-0.0029
AChE/BChE	Dual vs A-only	52	0.6815	0.8993	0.8948	0.6341	-0.0044
AChE/BChE	Dual vs B-only	55	0.7884	0.8201	0.8148	0.6442	-0.0053
F2/F10	Dual vs A-only	63	0.6139	0.9405	0.9335	0.4204	-0.0071
F2/F10	Dual vs B-only	63	0.4889	0.6633	0.6996	0.3740	0.0363
PPARG/PPARA	Dual vs A-only	63	0.7198	0.8206	0.8065	0.6764	-0.0141
PPARG/PPARA	Dual vs B-only	64	0.7158	0.6670	0.6611	0.6021	-0.0059
PPARA/PPARD	Dual vs A-only	64	0.4375	0.9316	0.9287	0.6787	-0.0029
PPARA/PPARD	Dual vs B-only	64	0.3115	0.8740	0.8789	0.5107	0.0049

All models and S1 are evaluated on identical members within each direction. The increment is AUROC(ECFP4 + S1) minus AUROC(ECFP4). Existing frozen five-fold assignments are retained; Q110 uses sorted canonical Murcko scaffolds assigned cyclically to folds 0-4. Standardization is fitted in training folds. Full descriptors, scaffolds, similarities, folds and OOF predictions are supplied as data files; no new model is fitted during table preparation. OOF refers to the learned chemistry models; S1 is a direct docking score.

Table S8 Random-seed and receptor-structure sensitivity

A. Five-seed distributions

Target pair	Metric	Seed 42	Median	Minimum	Maximum	IQR	Finite seeds
EGFR/HER2	AUROC B	0.6274	0.6375	0.6274	0.6477	0.0111	5
EGFR/HER2	AUROC A	0.2467	0.2388	0.2305	0.2467	0.0089	5
EGFR/HER2	Minimum AUROC	0.2467	0.2388	0.2305	0.2467	0.0089	5
JAK1/JAK2	AUROC B	0.5908	0.5840	0.5684	0.5908	0.0093	5
JAK1/JAK2	AUROC A	0.7485	0.7485	0.7422	0.7573	0.0112	5
JAK1/JAK2	Minimum AUROC	0.5908	0.5840	0.5684	0.5908	0.0093	5
JAK1/TYK2	AUROC B	0.6475	0.6475	0.6328	0.6631	0.0156	5
JAK1/TYK2	AUROC A	0.4219	0.4229	0.4121	0.4302	0.0015	5
JAK1/TYK2	Minimum AUROC	0.4219	0.4229	0.4121	0.4302	0.0015	5
PIK3CA/mTOR	AUROC B	0.7363	0.7354	0.6997	0.7402	0.0029	5
PIK3CA/mTOR	AUROC A	0.5801	0.5752	0.5596	0.5869	0.0190	5
PIK3CA/mTOR	Minimum AUROC	0.5801	0.5752	0.5596	0.5869	0.0190	5
AChE/BChE	AUROC B	0.6341	0.6422	0.6049	0.6533	0.0163	5
AChE/BChE	AUROC A	0.6442	0.6442	0.6356	0.6448	0.0026	5
AChE/BChE	Minimum AUROC	0.6341	0.6356	0.6049	0.6448	0.0075	5
F2/F10	AUROC B	0.4082	0.4062	0.3955	0.4082	0.0015	5
F2/F10	AUROC A	0.3623	0.3647	0.3594	0.3682	0.0029	5
F2/F10	Minimum AUROC	0.3623	0.3647	0.3594	0.3682	0.0029	5
PPARG/PPARA	AUROC B	0.6553	0.6553	0.6465	0.6680	0.0127	5
PPARG/PPARA	AUROC A	0.6021	0.6313	0.6021	0.6406	0.0225	5
PPARG/PPARA	Minimum AUROC	0.6021	0.6313	0.6021	0.6406	0.0225	5
PPARA/PPARD	AUROC B	0.6787	0.6929	0.6787	0.6953	0.0039	5
PPARA/PPARD	AUROC A	0.5107	0.5088	0.4800	0.5151	0.0146	5
PPARA/PPARD	Minimum AUROC	0.5107	0.5088	0.4800	0.5151	0.0146	5

Five Vina seeds: 17, 29, 42, 71 and 101. Each seed uses its own valid population. Receptor comparisons use paired common members and replace only one target. Full direction-specific estimates, confidence intervals and members are in the source data. PIK3CA/mTOR Q110 has no alternative-receptor calculation. AUROC, target B compares dual with A-only; AUROC, target A compares dual with B-only. Minimum AUROC is the smaller directional AUROC.

Table S8 Random-seed and receptor-structure sensitivity

B. One-receptor substitutions

Target pair	Alt. PDB	Side	n (dual / A-only / B-only)	Original minimum AUROC	Alternative minimum AUROC	Minimum AUROC difference (95% CI)	Affected AUROC difference	Unreplaced AUROC difference
AChE/BChE	1P0M	B	26 / 23 / 26	0.6037	0.6003	-0.0033 [-0.0803, 0.0836]	-0.0033	0
AChE/BChE	4EY6	A	27 / 23 / 26	0.6184	0.5670	-0.0514 [-0.1852, 0]	-0.0940	0
F2/F10	2Y5F	B	30 / 32 / 31	0.3828	0.3828	0 [-0.0469, 0.0229]	-0.0146	0
F2/F10	3SHC	A	30 / 32 / 31	0.3828	0.4333	0.0505 [0, 0.1441]	0.0780	0
PPARA/PPARD	5U46	B	32 / 32 / 32	0.5107	0.5107	0 [-0.0225, 0]	-0.0181	0
PPARA/PPARD	6KAX	A	32 / 32 / 32	0.5107	0.4438	-0.0669 [-0.1484, 0.0044]	-0.0669	0

Five Vina seeds: 17, 29, 42, 71 and 101. Each seed uses its own valid population. Receptor comparisons use paired common members and replace only one target. Full direction-specific estimates, confidence intervals and members are in the source data. PIK3CA/mTOR Q110 has no alternative-receptor calculation. AUROC, target B compares dual with A-only; AUROC, target A compares dual with B-only. Minimum AUROC is the smaller directional AUROC.

Table S9 Chemical-diversity and activity-threshold sensitivity

A. CAP5 membership

Target pair	Original n	CAP5 n	Excluded n	Excluded dual	Excluded A-only	Excluded B-only	Excluded neither
EGFR/HER2	109	109	0	0	0	0	0
JAK1/JAK2	110	107	3	0	3	0	0
JAK1/TYK2	110	110	0	0	0	0	0
PIK3CA/mTOR	110	110	0	0	0	0	0
AChE/BChE	98	98	0	0	0	0	0
F2/F10	110	110	0	0	0	0	0
PPARG/PPARA	110	110	0	0	0	0	0
PPARA/PPARD	110	107	3	0	1	2	0

CAP5 is post hoc supporting sensitivity: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S9 Chemical-diversity and activity-threshold sensitivity

B. Vina minimum AUROC before and after CAP5

Target pair	Primary minimum AUROC (95% CI)	CAP5 minimum AUROC (95% CI)	Point change
EGFR/HER2	0.2467 [0.1328, 0.3783]	0.2467 [0.1328, 0.3783]	0.0000
JAK1/JAK2	0.5908 [0.4512, 0.7246]	0.6315 [0.4871, 0.7597]	0.0406
JAK1/TYK2	0.4219 [0.2812, 0.5635]	0.4219 [0.2812, 0.5635]	0.0000
PIK3CA/mTOR	0.5801 [0.4346, 0.7129]	0.5801 [0.4346, 0.7129]	0.0000
AChE/BChE	0.6341 [0.4607, 0.7540]	0.6341 [0.4607, 0.7540]	0.0000
F2/F10	0.3623 [0.2305, 0.4805]	0.3623 [0.2305, 0.4805]	0.0000
PPARG/PPARA	0.6021 [0.4561, 0.7051]	0.6021 [0.4561, 0.7051]	0.0000
PPARA/PPARD	0.5107 [0.3652, 0.6475]	0.4969 [0.3427, 0.6354]	-0.0139

CAP5 is post hoc supporting sensitivity: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S9 Chemical-diversity and activity-threshold sensitivity

C. PIK3CA/mTOR strict 6.5/5.5 subset

Target pair	Scheme	n (dual / A-only / B-only)	AUROC, target B (95% CI)	AUROC, target A (95% CI)	Minimum AUROC (95% CI)
PIK3CA/mTOR	S1	25 / 12 / 14	0.7900 [0.6200, 0.9300]	0.4571 [0.2743, 0.6429]	0.4571 [0.2743, 0.6429]
PIK3CA/mTOR	S2	25 / 12 / 14	0.6200 [0.4133, 0.8133]	0.5343 [0.3429, 0.7229]	0.5343 [0.3343, 0.6829]
PIK3CA/mTOR	S3	25 / 12 / 14	0.7500 [0.5767, 0.8933]	0.6286 [0.4457, 0.7971]	0.6286 [0.4429, 0.7800]
PIK3CA/mTOR	S4	25 / 12 / 14	0.6267 [0.4267, 0.8167]	0.6057 [0.4143, 0.7829]	0.6057 [0.3867, 0.7229]
PIK3CA/mTOR	S5	25 / 12 / 14	0.5933 [0.4033, 0.7833]	0.6686 [0.4828, 0.8371]	0.5933 [0.4000, 0.7367]

CAP5 is post hoc supporting sensitivity: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S9 Chemical-diversity and activity-threshold sensitivity

D. Chemistry-model sensitivity to CAP5

Target pair	Direction	Original OOF n	CAP5 OOF n	Original ECFP4 AUROC	CAP5 ECFP4 AUROC	ECFP4 AUROC change	Original Vina increment	CAP5 Vina increment
EGFR/HER2	Dual vs A-only	65	65	0.8002	0.8002	0.0000	-0.0164	-0.0164
EGFR/HER2	Dual vs B-only	59	59	0.8790	0.8790	0.0000	-0.0023	-0.0023
JAK1/JAK2	Dual vs A-only	64	61	0.9185	0.9079	-0.0106	0.0015	0.0048
JAK1/JAK2	Dual vs B-only	64	64	0.9678	0.9678	0.0000	0.0029	0.0029
JAK1/TYK2	Dual vs A-only	63	63	0.8488	0.8488	0.0000	-0.0050	-0.0050
JAK1/TYK2	Dual vs B-only	63	63	0.9042	0.9042	0.0000	-0.0010	-0.0010
PIK3CA/mTOR	Dual vs A-only	64	64	0.8213	0.8213	0.0000	0.0215	0.0215
PIK3CA/mTOR	Dual vs B-only	64	64	0.7285	0.7285	0.0000	-0.0029	-0.0029
AChE/BChE	Dual vs A-only	52	52	0.8993	0.8993	0.0000	-0.0044	-0.0044
AChE/BChE	Dual vs B-only	55	55	0.8201	0.8201	0.0000	-0.0053	-0.0053
F2/F10	Dual vs A-only	63	63	0.9405	0.9405	0.0000	-0.0071	-0.0071
F2/F10	Dual vs B-only	63	63	0.6633	0.6633	0.0000	0.0363	0.0363
PPARG/PPARA	Dual vs A-only	63	63	0.8206	0.8206	0.0000	-0.0141	-0.0141
PPARG/PPARA	Dual vs B-only	64	64	0.6670	0.6670	0.0000	-0.0059	-0.0059
PPARA/PPARD	Dual vs A-only	64	63	0.9316	0.9345	0.0028	-0.0029	-0.0030
PPARA/PPARD	Dual vs B-only	64	62	0.8740	0.8688	-0.0053	0.0049	0.0042

CAP5 is post hoc supporting sensitivity: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S10 Joint-ranking composition and directional concordance

A1. Method-specific top-10% counts

Target pair	Scheme	N	k	Dual in population	Dual selected	A-only selected	B-only selected
EGFR/HER2	S1	97	10	28	1	4	5
EGFR/HER2	S2	97	10	28	2	1	7
EGFR/HER2	S3	97	10	28	2	4	4
EGFR/HER2	S4	97	10	28	2	2	6
EGFR/HER2	S5	97	10	28	2	0	8
JAK1/JAK2	S1	96	10	32	6	3	1
JAK1/JAK2	S2	96	10	32	7	3	0
JAK1/JAK2	S3	96	10	32	4	4	2
JAK1/JAK2	S4	96	10	32	4	3	3
JAK1/JAK2	S5	96	10	32	7	3	0
JAK1/TYK2	S1	96	10	32	3	1	6
JAK1/TYK2	S2	96	10	32	6	2	2
JAK1/TYK2	S3	96	10	32	5	0	5
JAK1/TYK2	S4	96	10	32	5	3	2
JAK1/TYK2	S5	96	10	32	5	4	1
PIK3CA/mTOR	S1	96	10	32	4	0	6
PIK3CA/mTOR	S2	96	10	32	6	3	1
PIK3CA/mTOR	S3	96	10	32	7	1	2
PIK3CA/mTOR	S4	96	10	32	5	3	2
PIK3CA/mTOR	S5	96	10	32	7	1	2
AChE/BChE	S1	80	8	27	6	0	2
AChE/BChE	S2	79	8	27	3	2	3
AChE/BChE	S3	80	8	27	3	3	2
AChE/BChE	S4	79	8	27	3	3	2
AChE/BChE	S5	82	9	27	3	1	5
F2/F10	S1	96	10	32	4	4	2
F2/F10	S2	95	10	31	4	3	3
F2/F10	S3	96	10	32	7	1	2
F2/F10	S4	95	10	31	5	5	0
F2/F10	S5	95	10	31	6	3	1

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S10 Joint-ranking composition and directional concordance (continued)

A1. Method-specific top-10% counts

Target pair	Scheme	N	k	Dual in population	Dual selected	A-only selected	B-only selected
PPARG/PPARA	S1	96	10	32	6	3	1
PPARG/PPARA	S2	94	10	30	2	4	4
PPARG/PPARA	S3	96	10	32	4	3	3
PPARG/PPARA	S4	94	10	30	3	4	3
PPARG/PPARA	S5	96	10	32	1	4	5
PPARA/PPARD	S1	96	10	32	3	2	5
PPARA/PPARD	S2	91	10	27	5	2	3
PPARA/PPARD	S3	96	10	32	6	2	2
PPARA/PPARD	S4	91	10	27	4	1	5
PPARA/PPARD	S5	96	10	32	5	1	4

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S10 Joint-ranking composition and directional concordance

A2. Method-specific top-10% fractions

Target pair	Scheme	Dual fraction	Single-target fraction	Dual recall	Population dual fraction
EGFR/HER2	S1	0.1000	0.9000	0.0357	0.2887
EGFR/HER2	S2	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S3	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S4	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S5	0.2000	0.8000	0.0714	0.2887
JAK1/JAK2	S1	0.6000	0.4000	0.1875	0.3333
JAK1/JAK2	S2	0.7000	0.3000	0.2188	0.3333
JAK1/JAK2	S3	0.4000	0.6000	0.1250	0.3333
JAK1/JAK2	S4	0.4000	0.6000	0.1250	0.3333
JAK1/JAK2	S5	0.7000	0.3000	0.2188	0.3333
JAK1/TYK2	S1	0.3000	0.7000	0.0938	0.3333
JAK1/TYK2	S2	0.6000	0.4000	0.1875	0.3333
JAK1/TYK2	S3	0.5000	0.5000	0.1562	0.3333
JAK1/TYK2	S4	0.5000	0.5000	0.1562	0.3333
JAK1/TYK2	S5	0.5000	0.5000	0.1562	0.3333
PIK3CA/mTOR	S1	0.4000	0.6000	0.1250	0.3333
PIK3CA/mTOR	S2	0.6000	0.4000	0.1875	0.3333
PIK3CA/mTOR	S3	0.7000	0.3000	0.2188	0.3333
PIK3CA/mTOR	S4	0.5000	0.5000	0.1562	0.3333
PIK3CA/mTOR	S5	0.7000	0.3000	0.2188	0.3333
AChE/BChE	S1	0.7500	0.2500	0.2222	0.3375
AChE/BChE	S2	0.3750	0.6250	0.1111	0.3418
AChE/BChE	S3	0.3750	0.6250	0.1111	0.3375
AChE/BChE	S4	0.3750	0.6250	0.1111	0.3418
AChE/BChE	S5	0.3333	0.6667	0.1111	0.3293
F2/F10	S1	0.4000	0.6000	0.1250	0.3333
F2/F10	S2	0.4000	0.6000	0.1290	0.3263
F2/F10	S3	0.7000	0.3000	0.2188	0.3333
F2/F10	S4	0.5000	0.5000	0.1613	0.3263
F2/F10	S5	0.6000	0.4000	0.1935	0.3263

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S10 Joint-ranking composition and directional concordance (continued)

A2. Method-specific top-10% fractions

Target pair	Scheme	Dual fraction	Single-target fraction	Dual recall	Population dual fraction
PPARG/PPARA	S1	0.6000	0.4000	0.1875	0.3333
PPARG/PPARA	S2	0.2000	0.8000	0.0667	0.3191
PPARG/PPARA	S3	0.4000	0.6000	0.1250	0.3333
PPARG/PPARA	S4	0.3000	0.7000	0.1000	0.3191
PPARG/PPARA	S5	0.1000	0.9000	0.0312	0.3333
PPARA/PPARD	S1	0.3000	0.7000	0.0938	0.3333
PPARA/PPARD	S2	0.5000	0.5000	0.1852	0.2967
PPARA/PPARD	S3	0.6000	0.4000	0.1875	0.3333
PPARA/PPARD	S4	0.4000	0.6000	0.1481	0.2967
PPARA/PPARD	S5	0.5000	0.5000	0.1562	0.3333

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S10 Joint-ranking composition and directional concordance

B. Paired changes versus Vina

Target pair	Scheme vs S1	Ranking common n	Directional common dual n	Minimum AUROC difference	Single-target fraction difference	Concordance	Supporting comparison
EGFR/HER2	S2	97	28	-0.0257	-0.1000	Discordant	NO
EGFR/HER2	S3	97	28	0.0167	-0.1000	Concordant improvement	NO
EGFR/HER2	S4	97	28	0.0346	-0.1000	Concordant improvement	YES
EGFR/HER2	S5	97	28	-0.0011	-0.1000	Discordant	NO
JAK1/JAK2	S2	96	32	-0.1523	-0.1000	Discordant	NO
JAK1/JAK2	S3	96	32	-0.1172	0.2000	Concordant worsening	NO
JAK1/JAK2	S4	96	32	-0.1465	0.2000	Concordant worsening	YES
JAK1/JAK2	S5	96	32	-0.0771	-0.1000	Discordant	NO
JAK1/TYK2	S2	96	32	0.1377	-0.3000	Concordant improvement	NO
JAK1/TYK2	S3	96	32	0.0654	-0.2000	Concordant improvement	NO
JAK1/TYK2	S4	96	32	0.2539	-0.2000	Concordant improvement	YES
JAK1/TYK2	S5	96	32	0.2471	-0.2000	Concordant improvement	NO
PIK3CA/mTOR	S2	96	32	-0.0049	-0.2000	Discordant	NO
PIK3CA/mTOR	S3	96	32	0.1240	-0.3000	Concordant improvement	NO
PIK3CA/mTOR	S4	96	32	-0.0635	-0.1000	Discordant	YES
PIK3CA/mTOR	S5	96	32	0.0176	-0.3000	Concordant improvement	NO
AChE/BChE	S2	79	27	-0.1416	0.3750	Concordant worsening	NO
AChE/BChE	S3	80	27	-0.1433	0.3750	Concordant worsening	NO
AChE/BChE	S4	79	27	-0.2431	0.3750	Concordant worsening	YES
AChE/BChE	S5	80	27	-0.1830	0.3750	Concordant worsening	NO

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S10 Joint-ranking composition and directional concordance (continued)

B. Paired changes versus Vina

Target pair	Scheme vs S1	Ranking common n	Directional common dual n	Minimum AUROC difference	Single-target fraction difference	Concordance	Supporting comparison
F2/F10	S2	95	31	0.1643	0	No change	NO
F2/F10	S3	96	32	0.2471	-0.3000	Concordant improvement	NO
F2/F10	S4	95	31	0.1956	-0.1000	Concordant improvement	YES
F2/F10	S5	95	31	0.1815	-0.2000	Concordant improvement	NO
PPARG/PPARA	S2	94	30	-0.3089	0.4000	Concordant worsening	NO
PPARG/PPARA	S3	96	32	-0.2915	0.2000	Concordant worsening	NO
PPARG/PPARA	S4	94	30	-0.2578	0.3000	Concordant worsening	YES
PPARG/PPARA	S5	96	32	-0.1558	0.5000	Concordant worsening	NO
PPARA/PPARD	S2	91	27	0.1192	-0.2000	Concordant improvement	NO
PPARA/PPARD	S3	96	32	0.1582	-0.3000	Concordant improvement	NO
PPARA/PPARD	S4	91	27	0	-0.1000	No change	YES
PPARA/PPARD	S5	96	32	0.1230	-0.2000	Concordant improvement	NO

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S10 Joint-ranking composition and directional concordance

C. Concordance counts

Scheme vs S1	Pairs	Concordant improvement	Concordant worsening	Discordant	No change
S2	8	2	2	3	1
S3	8	5	3	0	0
S4	8	3	3	1	1
S5	8	4	2	2	0

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S11. Exploratory single-target controls

A. Single-target-active versus neither

Target pair	Score target	Positive class	n positive	n negative	AUROC (95% CI)
EGFR/HER2	EGFR	A-only	37	12	0.5766 [0.3761, 0.7680]
EGFR/HER2	HER2	B-only	32	12	0.8672 [0.7109, 0.9818]
JAK1/JAK2	JAK1	A-only	32	14	0.6049 [0.4241, 0.7790]
JAK1/JAK2	JAK2	B-only	32	14	0.5446 [0.3504, 0.7388]
JAK1/TYK2	JAK1	A-only	32	14	0.7455 [0.5558, 0.9085]
JAK1/TYK2	TYK2	B-only	32	14	0.7522 [0.5804, 0.8996]
PIK3CA/mTOR	PIK3CA	A-only	32	14	0.5357 [0.3549, 0.7121]
PIK3CA/mTOR	mTOR	B-only	32	14	0.6183 [0.4576, 0.7768]
AChE/BChE	AChE	A-only	25	15	0.6213 [0.4427, 0.7867]
AChE/BChE	BChE	B-only	28	15	0.6071 [0.4310, 0.7786]
F2/F10	F2	A-only	32	14	0.6339 [0.4621, 0.7879]
F2/F10	F10	B-only	32	14	0.6406 [0.4353, 0.8304]
PPARG/PPARA	PPARG	A-only	32	14	0.6685 [0.4631, 0.8504]
PPARG/PPARA	PPARA	B-only	32	14	0.6384 [0.4330, 0.8281]
PPARA/PPARD	PPARA	A-only	32	14	0.5223 [0.3303, 0.7143]
PPARA/PPARD	PPARD	B-only	32	14	0.7009 [0.5290, 0.8527]

S1: negative Vina docking energy, with higher values favorable (search seed 42). The positive class is the single-target class active at the scored target. Neither ligands are negative in panel A; the opposite single-target class is negative in panel B. Dual ligands are excluded. Counts include eligible observations with a valid score at that target. CI: 95% class-stratified percentile bootstrap, 10,000 resamples, seed 271828; linear 2.5th and 97.5th percentiles. Positive resamples are shared between the two comparisons. Score ties count as 0.5. Intervals are exploratory and unadjusted for multiplicity.

Table S11. Exploratory single-target controls

B. Single-target-active versus opposite single-target-active

Target pair	Score target	Positive class	n positive	n negative	AUROC (95% CI)	
EGFR/HER2	EGFR	A-only	37	32	0.1807 [0.0845, 0.2914]	
EGFR/HER2	HER2	B-only	32	37	0.7939 [0.6748, 0.8970]	
JAK1/JAK2	JAK1	A-only	32	32	0.6250 [0.4844, 0.7588]	
JAK1/JAK2	JAK2	B-only	32	32	0.3755 [0.2417, 0.5186]	
JAK1/TYK2	JAK1	A-only	32	32	0.3125 [0.1865, 0.4512]	
JAK1/TYK2	TYK2	B-only	32	32	0.6777 [0.5410, 0.8057]	
PIK3CA/mTOR	PIK3CA	A-only	32	32	0.4268 [0.2900, 0.5713]	
PIK3CA/mTOR	mTOR	B-only	32	32	0.6348 [0.4941, 0.7686]	
AChE/BChE	AChE	A-only	25	28	0.7000 [0.5500, 0.8286]	
AChE/BChE	BChE	B-only	28	25	0.5293 [0.3629, 0.6900]	
F2/F10	F2	A-only	32	32	0.4619 [0.3213, 0.6084]	
F2/F10	F10	B-only	32	32	0.5039 [0.3584, 0.6445]	
PPARG/PPARA	PPARG	A-only	32	32	0.5044 [0.3584, 0.6475]	
PPARG/PPARA	PPARA	B-only	32	32	0.5083 [0.3623, 0.6523]	
PPARA/PPARD	PPARA	A-only	32	32	0.5542 [0.4102, 0.6958]	
PPARA/PPARD	PPARD	B-only	32	32	0.7144 [0.5801, 0.8345]	

S1: negative Vina docking energy, with higher values favorable (search seed 42). The positive class is the single-target class active at the scored target. Neither ligands are negative in panel A; the opposite single-target class is negative in panel B. Dual ligands are excluded. Counts include eligible observations with a valid score at that target. CI: 95% class-stratified percentile bootstrap, 10,000 resamples, seed 271828; linear 2.5th and 97.5th percentiles. Positive resamples are shared between the two comparisons. Score ties count as 0.5. Intervals are exploratory and unadjusted for multiplicity.